LIBSEAT

Library Seat & Resource Management System

BITS Pilani Hyderabad · Product Requirements Document · PM Portfolio

PostgreSQL Backend

Flask REST API

Student + Librarian Portal

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THE PROBLEM

"Students at BITS Pilani Hyderabad spend valuable time hunting for library seats and reference books with no centralised visibility into availability — leading to wasted trips, double-bookings, and frustrated librarians."

No Seat Visibility

Students walk to the library only to find all seats occupied; zero real-time availability data

Book Conflicts

Multiple students reserve the same reference book for overlapping time slots causing disputes

Manual Overhead

Librarians manually manage holiday cancellations, seat reassignments, and daily booking logs

USER PERSONAS

Aryan, 2nd Year CS

Student

Goals

- Book a seat + reference book together
- Check real-time seat availability
- See booking history

Pain Points

- Wasted trips to full library
- Can't reserve books in advance
- No group seating coordination

Dr. Priya

Librarian

Goals

- Monitor seat occupancy by room
- Declare holidays & auto-cancel
- Generate daily usage reports

Pain Points

- Manual cancellation on holidays
- No live dashboard
- Book conflicts with same ISBN

Srinivas

System Admin

Goals

- Add/remove seats and books
- Manage student records
- Ensure data integrity via triggers

Pain Points

- Cascading conflicts on seat removal
- No automated reassignment
- Manual email validation

SYSTEM OVERVIEW

PostgreSQL

Primary Database
Triggers & Procedures

Flask (Python)

REST API Backend
Route Handlers

Web Frontend

Student Portal
Librarian Dashboard

Auth Layer

Email Validation
@hyderabad.bits-pilani.ac.in

Core Data Model

Student	Seat	Book	Booking	BookingBook_ID
student_id (PK)	seat_id (PK)	b_id (PK)	student_id (FK)	student_id (FK)
name, email	seat_no, location	isbn, b_name	seat_id (FK)	book_id (FK)
@bits email enforced	managed by trigger	authors, category	start_time, end_time	start_time, end_time

CORE FEATURES — BOOKING FLOW

Seat Availability Check

P0 · MVP

/available-seats

Real-time query returns all seats not booked during the requested time window

Book Availability Check

P0 · MVP

/available-books

Returns books not currently assigned to any booking overlapping the requested slot

Conflict Detection

P0 · MVP

/check-booking-conflict

PostgreSQL function validates no time-overlap exists for the student before confirming booking

Booking History

P0 · MVP

/booking-history

Full join across booking, seat, and book tables returns paginated history for any student

Current Booking

P1

/current-booking

Retrieves the student's most recent active booking with seat and book details

Holiday Declaration

P1

Stored Procedure

Librarian triggers procedure that deletes all next-day bookings when holiday is declared

TRIGGERS & AUTOMATION

🔄 Seat Deletion → Auto-Reassign

BEFORE DELETE ON seat

T-01 — Logic Flow:

1. Admin deletes a seat
2. Trigger fires BEFORE DELETE
3. Finds next available free seat
4. Reassigns all affected bookings
5. Cancels bookings with no alternative

📚 Book Deletion → ISBN Match

BEFORE DELETE ON book

T-02 — Logic Flow:

1. Librarian removes a book
2. Trigger fires BEFORE DELETE
3. Finds same-ISBN copy that's free
4. Reassigns all affected BookingBook_IDs
5. Cancels booking-books with no copy

Both triggers follow a Try-Reassign-Then-Cancel pattern — zero data loss, maximum service continuity

LIBRARIAN DASHBOARD & ANALYTICS

Hourly Occupancy Stats

At each 1-hour interval, shows total seats occupied across all locations — enables librarians to identify peak usage windows and plan staffing

Room Occupancy %

Percentage of seats occupied per room/location at each hour — identifies underutilised spaces and bottleneck rooms for capacity planning

⚠ Last Seat Alert

Trigger-driven alert fires when only 1 seat remains in a given location — displayed on dashboard and optionally sent to the waiting list queue

Holiday Management

Stored procedure deletes all next-day bookings in a single atomic transaction when librarian declares a holiday — with audit log entry

SUCCESS METRICS & KPIs

Student Experience

Seat Booking Completion Rate	**> 90% successful bookings**
Avg. Time to Find Available Seat	**< 30 seconds**
Double-Booking Incidents	**0 conflicts post-launch**

Librarian Efficiency

Holiday Cancellation Time	**< 10 seconds (vs. hours manually)**
Manual Intervention Rate	**< 5% of daily bookings**
Dashboard Load Time	**< 2 seconds for hourly stats**

System Health

Trigger Execution Success Rate	**> 99.9%**
Seat Reassignment Rate on Deletion	**> 80% auto-resolved**
Database Query Response Time	**< 500ms for availability queries**

Adoption

Weekly Active Students	**> 60% of enrolled using portal**
Repeat Booking Rate	**> 70% of users book again in 7 days**
Librarian Dashboard DAU	**100% of librarians daily**

TECHNICAL IMPLEMENTATION

⚡ PostgreSQL Functions

`check_booking_conflict()`

Returns BOOLEAN — checks if student has overlapping booking. Called before every INSERT into booking table.

`get_seat_count()`

Returns INT count of seats in a given location. Used internally by get_next_available_seat().

`get_next_available_seat()`

Uses generate_series() to find the lowest-numbered unoccupied seat_no in a location. Supports auto-assignment.

API Endpoints

GET	`/available-seats`	Check free seats for time window
GET	`/available-books`	Check free books for time window
GET	`/check-booking-conflict`	Validate no overlap before booking
GET	`/booking-history`	Full history for a student
GET	`/current-booking`	Active booking for a student
POST	`/declare-holiday`	Trigger holiday procedure (Librarian)

PRODUCT ROADMAP

Phase 1 Week 1–3

Core Booking System

- Student registration with email validation
- Seat availability API
- Book availability API
- Conflict detection function
- Basic booking creation flow

✓ Validate core booking + conflict detection

Phase 2 Week 4–6

Automation & Triggers

- Seat deletion trigger (auto-reassign)
- Book deletion trigger (ISBN match)
- Holiday declaration procedure
- Last-seat alert trigger
- Booking history & current booking APIs

✓ Zero-touch seat & book reassignment

Phase 3 Week 7–8

Analytics & Dashboard

- Hourly occupancy stats query
- Room-level occupancy % report
- Librarian web dashboard
- Export booking reports (CSV)
- Mobile-responsive student portal

✓ Librarian intelligence layer

WHY LIBSEAT WORKS

Resilient by Design

BEFORE DELETE triggers ensure no student ever loses a booking without an alternative — the system degrades gracefully, not catastrophically.

Sub-Second Availability

Optimised NOT IN subqueries with time-overlap logic deliver real-time seat and book availability without full table scans.

Domain-Specific Auth

Email validation enforced at the procedure level ensures only legitimate BITS Pilani Hyderabad students can register — no application-layer loophole.

Built-in Analytics

Hourly occupancy and room utilisation queries are native SQL — no external BI tool needed for the librarian to make data-driven decisions.